STEPHEN NJUGUNA

Artificial Intelligence | Data Science | Computer Science | Digital Innovation

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PROFESSIONAL PROFILE

Computer Science graduate and applied AI/data practitioner with a strong interest in using artificial intelligence, data science and digital systems to address practical development challenges. Academic training at Rongo University (2019–2023) is complemented by professional learning in AI and data science, including the IBM Professional Data Science pathway. Builds and experiments with machine-learning, generative-AI, reinforcement-learning and data-driven applications, with a particular focus on education, TVET, institutional analytics and digital transformation. Demonstrated ability to move from a real-world problem to data preparation, analysis, software implementation, visualization and deployable digital products.

CORE VALUE PROPOSITION

- **Applied AI:** builds practical AI-enabled applications rather than treating AI as theory alone; experience spans GenAI, AI agents and reinforcement learning.
- **Data-to-decision:** turns survey and institutional data into dashboards, indicators and decision-support outputs.
- **Development orientation:** repeatedly applies technology to education, TVET, institutional management, agriculture, employment and African information-access contexts.
- **Builder mindset:** combines Computer Science fundamentals with rapid prototyping, modern AI-assisted development and iterative testing.
- **Responsible innovation:** interested in reliability, validation, risk controls, human oversight and the social implications of AI deployment.

SELECTED AI, DATA SCIENCE & INNOVATION PROJECTS

Trading AI – Reinforcement Learning Intelligence Platform Python, RL, FastAPI, PyTorch, Stable-Baselines3

- Developed a research-oriented platform combining reinforcement learning, market data, backtesting, risk management, experiments and TradingView integration.
- Implemented an API-driven architecture with health monitoring, dashboard summaries, backtesting and RL-training endpoints; project documentation records automated testing and explicit safety controls.

IBM Professional Data Science Project Portfolio Python, Data Science, ML

- Completed and documented practical work associated with the IBM Professional Data Science Certificate, establishing a formal portfolio in data analysis and applied data science.

Gen-AI Agents Notebook Training Generative AI, AI Agents

- Developed practical learning and experimentation materials around generative AI and agent-based workflows, strengthening hands-on understanding of emerging AI application patterns.

AI Debate Arena AI Application Development

- Built an AI-focused application exploring interactive, model-driven debate and reasoning experiences.

Disease Prediction Model Machine Learning

- Implemented a machine-learning prediction project as part of applied data/AI learning, demonstrating supervised modelling and practical predictive analytics.

DIGITAL TRANSFORMATION & SYSTEMS PROJECTS

- **Kussala Institute / St John's Websites:** designed and developed institutional web platforms, combining user-facing information architecture with modern web development.
- **Farm Management for Livestock System:** applied software engineering to an agricultural management use case, demonstrating cross-domain problem solving.
- **Automated Mathematical Research Platform:** explored automation for computational research and mathematical workflows.

TECHNICAL SKILLS

Programming: Python, JavaScript, PHP, HTML/CSS, SQL
Data Science: Pandas, NumPy, scikit-learn, exploratory analysis, data cleaning, statistical analysis, visualization
AI/ML: Generative AI, LLM applications, AI agents, machine learning, reinforcement learning, model experimentation
Data Apps: Streamlit, Plotly, interactive dashboards, KPI/reporting systems
Backend/Web: Flask, Django, FastAPI, Vue.js, REST APIs
Databases: MySQL, SQLite, PostgreSQL

Education Technology: Moodle, WordPress, LMS design and digital learning systems
Tools: Git, GitHub, Docker, Anaconda, Jupyter, AI-assisted development environments

EDUCATION

Rongo University, Kenya 2019–2023
Computer Science

Computer Science foundation spanning programming, databases, software development, systems analysis and computational problem solving, followed by continued specialization through professional AI and data-science learning.

CERTIFICATIONS & CONTINUOUS LEARNING

- **IBM Professional Data Science Certificate** – Coursera/IBM; practical portfolio maintained on GitHub.
- **Additional AI and emerging-technology certifications** – multiple professional courses/certificates in artificial intelligence and related technologies.
- Continuous self-directed learning in Generative AI, AI agents, machine learning, reinforcement learning, data science and digital innovation.

INNOVATION INTERESTS

AI for Development • AI for Education and TVET • Data-driven public/institutional decision making • Responsible AI • Digital transformation • Youth and skills development • African technology ecosystems

CAREER OBJECTIVE

To contribute to an innovation-focused team where AI, data science and digital technologies are translated into measurable development impact. I am particularly motivated by opportunities that connect technical innovation with education, skills, institutional effectiveness, inclusive digital transformation and evidence-based decision making in African contexts.

ONLINE PORTFOLIO

github.com/devstephenke-sys Selected repositories include AI, data science, reinforcement learning, TVET and institutional technology projects.

References available upon request.